

AI RESUME ANALYZER AND JOB RECOMMENDATION SYSTEM

Major Project 2 — Project Report

NLP-Based Resume Analysis, Job Matching and Learning Roadmap

Python • Streamlit • Scikit-learn • Pandas • PyPDF • python-docx

1. Abstract

The AI Resume Analyzer and Job Recommendation System is an NLP-based application designed to analyze resumes and provide career-oriented guidance. The system accepts PDF or DOCX resumes, extracts and cleans the text, identifies job-related skills, compares the resume against predefined job-role profiles, calculates estimated match scores, recommends the top three roles, identifies missing skills for a selected target role, and generates a learning roadmap.

The implementation uses keyword-based skill extraction together with TF-IDF vectorization, cosine similarity, and skill-overlap analysis. A Streamlit dashboard provides an interactive interface for uploading a resume and viewing the analysis.

2. Problem Statement

Students and early-career candidates may find it difficult to determine which job roles align with their current technical skills and which skills they should develop next. This project addresses that problem by providing an automated resume-analysis workflow that connects extracted resume skills to predefined job-role requirements and produces an estimated role match and learning roadmap.

3. Objectives

- Upload and validate PDF or DOCX resumes.
- Extract readable resume text.
- Clean and normalize extracted text.
- Identify technical skills using a controlled skill dictionary.
- Compare resume content with predefined job-role profiles.
- Calculate estimated match scores.
- Recommend the top three job roles.
- Identify matched and missing skills for a selected target role.
- Generate a step-by-step learning roadmap.
- Provide the complete analysis through a Streamlit dashboard.

4. Technologies and Tools

Technology / Library	Purpose	Use in Project
Python	Programming language	Core implementation
Streamlit	Web interface	Interactive dashboard
PyPDF	PDF processing	PDF text extraction
python-docx	DOCX processing	DOCX text extraction
Pandas	Data handling	CSV datasets and results
NumPy	Numerical computing	Supporting data operations
Scikit-learn	ML / NLP utilities	TF-IDF and cosine similarity
Regex	Pattern matching	Skill detection and text cleaning

5. System Workflow

The project follows the workflow: resume upload → text extraction → text cleaning → skill identification → loading job-role requirements → comparison and scoring → role recommendation → skill-gap analysis → learning roadmap.

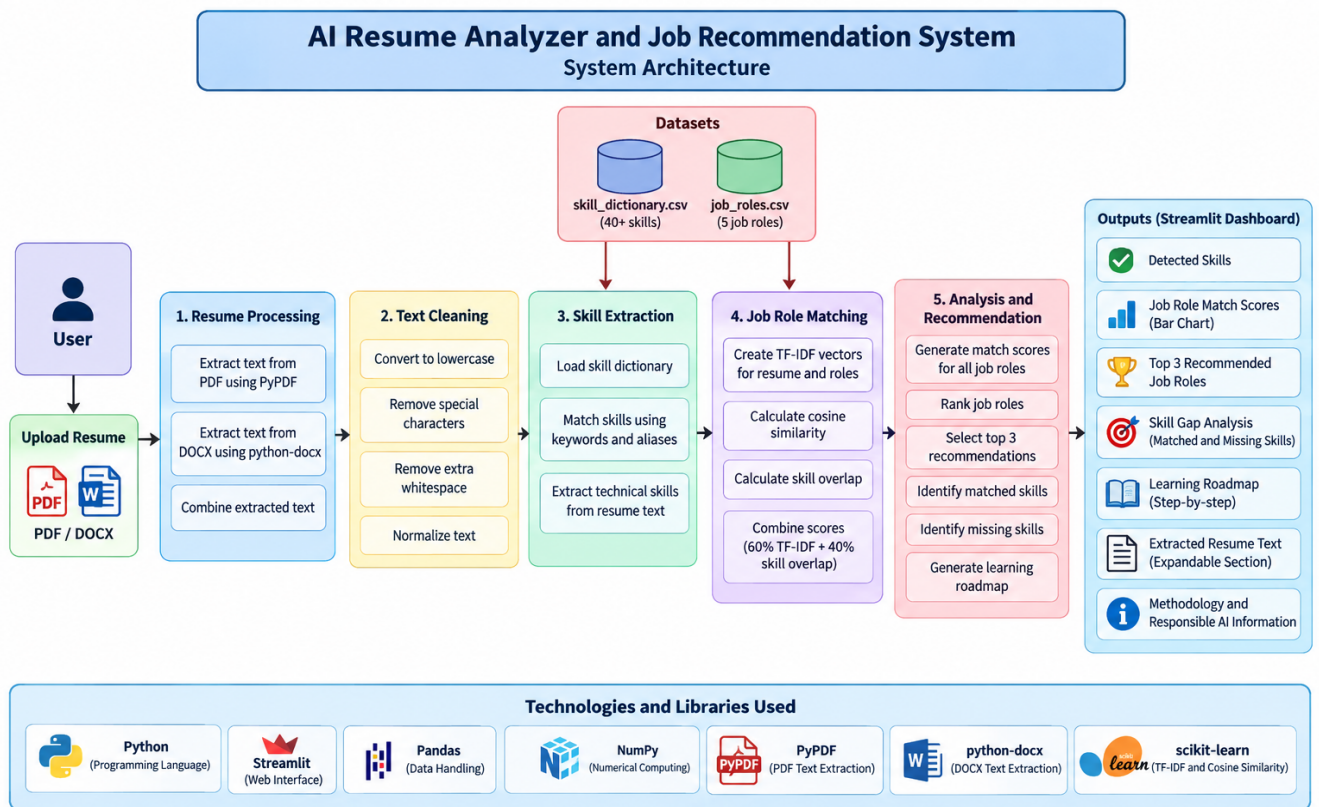


Figure 1. System architecture and workflow.

6. System Architecture

The application is organized into a Streamlit presentation layer and a consolidated backend engine. The Streamlit interface is implemented in `app.py`, while resume extraction, text cleaning, skill extraction, job-role loading, similarity calculation, role analysis, and roadmap generation are implemented in `resume_engine.py`.

Component	Responsibility
<code>app.py</code>	Streamlit user interface and dashboard.
<code>resume_engine.py</code>	Resume extraction, preprocessing, skill extraction, matching, recommendations and roadmap logic.
<code>data/skill_dictionary.csv</code>	Controlled list of 40+ technical skills and aliases.
<code>data/job_roles.csv</code>	Five predefined job roles with descriptions, required skills and roadmap steps.
<code>sample_resumes/</code>	Sample resume used for application testing.
<code>tests/test_cases.csv</code>	25 functional test cases.
<code>README.md</code>	Project documentation and setup instructions.
<code>requirements.txt</code>	Python dependency list.

7. Functional Modules

Resume Upload and Extraction: Accepts PDF and DOCX files and extracts readable text in memory.

Text Cleaning: Converts text to lowercase, normalizes whitespace, and prepares content for matching.

Skill Extraction: Checks resume text against a controlled dictionary of skills and aliases using pattern matching.

Job-Role Dataset: Loads five predefined roles with descriptions, required skills and roadmap steps.

Role Matching: Combines TF-IDF/cosine similarity with skill overlap to produce an estimated match score.

Recommendations: Ranks analyzed roles and displays the top three recommendations.

Skill-Gap Analysis: Shows skills detected in the resume and required skills that were not detected for the selected role.

Learning Roadmap: Combines missing skills with predefined role-specific learning steps.

8. Job Roles and Skill Dataset

The implementation contains five job roles:

- Data Analyst
- Machine Learning Engineer
- AI Engineer
- NLP Engineer
- Computer Vision Engineer

The skill dictionary contains more than the minimum 20–30 job-related skills specified for the project. It includes programming, databases, data analysis, machine learning, deep learning, NLP, computer vision, APIs, cloud and development tools.

9. Matching Methodology

9.1 TF-IDF and Cosine Similarity

The resume text and each job-role description are transformed into TF-IDF vectors. Cosine similarity is calculated to estimate how closely the resume content matches the role description.

9.2 Skill Overlap

The extracted resume skills are compared with the required skills of each role. The skill score is the number of matched required skills divided by the total number of required skills.

9.3 Final Estimated Match Score

The implementation combines the two components using a 60% TF-IDF similarity weight and a 40% skill-overlap weight. The result is converted to an estimated percentage and used to rank the predefined roles.

10. Streamlit Dashboard

- Resume upload for PDF/DOCX files.
- Detected skills displayed in the interface.
- Job-role match score bar chart.
- Top three recommended job roles.
- Target-role selection.
- Matched and missing skills.
- Suggested learning roadmap.
- Expandable extracted resume text.
- Methodology and responsible-AI information.

11. Results and Demonstration

The completed application was tested using a sample resume containing technical skills across data analysis, machine learning, AI, software development and related areas. The application successfully completed the end-to-end workflow and displayed detected skills, role match scores, top recommendations, skill gaps and a learning roadmap.

- Detected Skills
- Job Role Match Scores
- Top 3 Recommended Job Roles
- Detailed Skill Gap Analysis
- Suggested Learning Roadmap

12. Testing and Evaluation

A testing sheet containing 25 test cases was prepared. The cases cover file upload, text extraction, skill detection, role matching, recommendations, skill-gap analysis, roadmap generation, error handling and the complete workflow.

Test Area	Coverage
File handling	PDF, DOCX and unsupported file formats
Extraction	Readable resume text and unreadable/empty input
Skill extraction	Individual and multiple technical skills
Matching	Five available job roles and score generation
Recommendations	Top three role ranking
Skill gaps	Matched and missing skills
Roadmap	Learning steps for selected roles
Consistency	Repeated analysis and complete workflow

13. Responsible AI Considerations

The system is intended as a career-guidance and resume-analysis tool, not as an automatic hiring or rejection system. The implementation focuses on job-related skills and role requirements. It does not intentionally score gender, age, religion, nationality, photograph, marital status or disability. Match scores are estimates, and the absence of a detected keyword does not necessarily mean that a candidate lacks the underlying ability.

14. Limitations

- The skill extractor depends on the controlled skill dictionary and aliases.
- Skills not represented in the dictionary may not be detected.
- The system uses five predefined job roles rather than a live job-market dataset.
- Match scores are estimates based on resume content and predefined role profiles.
- Resume extraction quality depends on the source document being text-readable.
- The system is not intended to make hiring decisions.

15. Future Enhancements

- Advanced NLP with spaCy.
- Sentence Transformer embeddings.
- LLM-based resume feedback.

- Expansion of job roles and skill datasets.
- Integration with live job-market data.
- Downloadable analysis reports.
- FastAPI backend and database support.
- Docker and cloud deployment.

16. Conclusion

The AI Resume Analyzer and Job Recommendation System demonstrates a complete NLP-based workflow for resume analysis and career guidance. By combining resume text extraction, controlled skill identification, TF-IDF similarity, cosine similarity, skill overlap, role ranking and rule-based roadmap generation, the system provides an interactive way for users to understand their alignment with predefined job roles and identify areas for further learning.

17. Project Guidance Basis

This report follows the supplied Major Project 2 guidance, including its stated objectives, workflow, suggested tools, minimum feature requirements, folder structure, matching approaches, testing considerations, responsible-AI guidance and final deliverables.